INCH-POUND
MIL-PRF-27/19D
28 April 2008
SUPERSEDING
MIL-PRF-27/19C
15 April 1987

PERFORMANCE SPECIFICATION SHEET

TRANSFORMER, POWER 52 VA, 60 HZ

Inactive for New Design
after 6 June 1997

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-27.

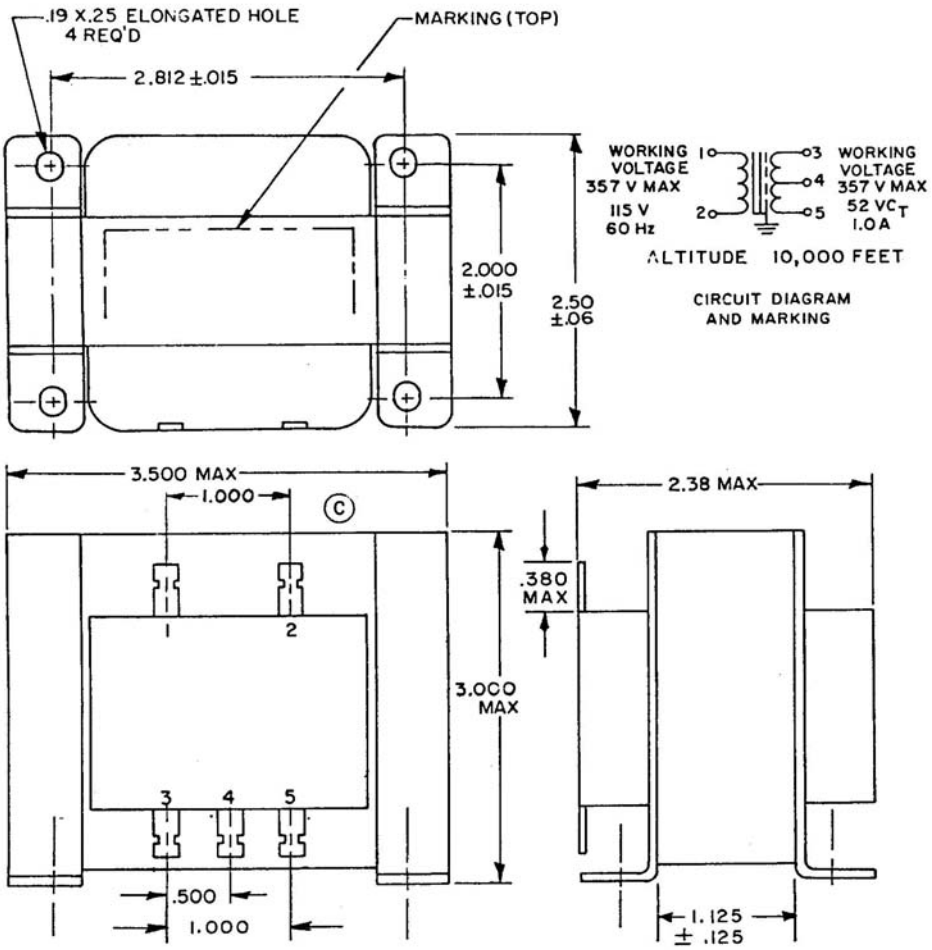


FIGURE 1. Dimensions and configuration.

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Inches	mm	Inches	mm
.015	0.38	1.000	25.40
.06	1.5	1.125	28.58
.125	3.18	2.000	50.80
.19	4.8	2.38	60.5
.25	6.4	2.50	63.5
.380	9.65	2.812	71.42
.500	12.70	3.000	76.20
		3.500	88.90

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.02 (0.5 mm) for two place decimals.
4. Tolerance on terminal position dimensions is ±.062 (1.57 mm).
5. Marking shall be on the top of the case.

FIGURE 1. Dimensions and configurations - Continued.

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REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the Extreme terminals of the winding.)

Electrical ratings:

Primary voltage (1-2): 58 voltamperes, 115 volts rms, 60 Hz.

Secondary voltage (3-5): 52 volts rms CT, 1.0 ampere rms.

Working voltage: (1-2) – 357 volts rms maximum.
(3 -5) – 357 volts rms maximum.

Design, configuration, and physical dimensions: See figure 1.

Duty cycle: Continuous.

Case: Encapsulated.
Material: Epoxy.

Terminals: Solder type.

Height: .38 inch maximum.

Weight: 3.5 pounds maximum.

Operating temperature: -55°C to +130°C

Altitude: 10,000 feet, maximum.

Terminal strength: Method 211 of MIL-STD-202, test condition A, 2 pounds.

Dielectric withstanding voltage (each winding) at sea level: Method 301 of MIL-STD-202; test voltage 1,000 volts rms.

Electrical characteristics:

No load: With 115 V rms and 60 Hz across (1-2), current in (1-2) shall not exceed 200 milliamperes, and the power in (1-2) shall not exceed 4.5 watts. Voltage across (3-5) shall be 55..5 volts rms ±2 percent.

Rated load: With 115 volts rms and 60 Hz across (1-2), and the rated current in secondary, voltage across (3-5) shall be 52 volts rms ±3 percent.

DC resistance:

(1-2): 9.89 ohms maximum.
(3-5): 1.43 ohms maximum.

Efficiency: 90 percent at rated load.

Regulation: $\frac{\text{Voltage (no load)} - \text{Voltage (rated load)}}{\text{Voltage (rated load}} \times 100$

Shall not exceed 12 percent

Polarity - Additive, with terminals 2 and 3 connected.

Electrostatic shielding: The voltage ratio shall me a minimum of 5 to 1 at 20 kHz.

Temperature rise: 40°C with 115 volts rms, 60 Hz across (1-2) at an ambient temperature of 90°C, full load terminals (3-5).

Marking location: See figure 1.

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Part or Identifying Number (PIN): M27/19-01.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents. In addition to MIL-PRF-27, this document references the following:

MIL-STD-202

Custodians:
Army - CR
Navy - EC
Air force – 11
DLA - CC

Preparing activity:
DLA - CC

Review activities:
Army – AR, CR4
Navy – AS, MC, OS, SH
Air Force – 19, 99

(Project: 5950-2008-003)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of this information above using the ASSIST Online database at http://assist.daps.dla.mil.

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